SUWAJ ARYAL

Ph.D. Applicant |M.Sc. Hydropower Engineering |B.E. Civil Engineering
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RESEARCH INTERESTS

Researcher specializing at the intersection of Water Resources Engineering and Artificial Intelligence. Focused on developing physics-informed machine learning frameworks, data-driven forecasting systems, and smart hydro-infrastructure solutions to tackle climate uncertainty and water security challenges..

- **Core Domains**: Computational Hydrology • Hydroinformatics • Climate Change Impact Modeling • Surface Water Dynamics
- **Methodologies**: Physics-Guided Deep Learning • Remote Sensing & GIS Analytics • Spatiotemporal Forecasting • Uncertainty Quantification
- **Applications**: Extreme-event prediction (Floods/Droughts) • Sustainable Hydropower Systems • Watershed Management • Smart Water Infrastructure • Flood Hazard Warning and mitigation

EDUCATION

M.Sc. in Hydropower Engineering *2024 – 2026*
Pulchowk Campus, Institute of Engineering, Tribhuvan University, Nepal
- Aggregate Percentage: 86% (Under top 5 percentile)

B.E. in Civil Engineering *2018 – 2023*
Pulchowk Campus, Institute of Engineering, Tribhuvan University, Nepal
- Aggregate Percentage: 72.5% (Top 20 percentile)

HONORS AND SCHOLARSHIPS

Fully funded M.Sc. Scholarship
- Awarded fully funded government scholarship for graduate studies based on top tier placement in national engineering competitive exam.

Undergraduate Merit Scholarship
- Secured highly competitive, fully funded government seat for Bachelor's program via the nationwide IOE entrance Examination.

Mahatma Gandhi Scholarship (Embassy of India, Kathmandu)
- A highly selective, merit-based government scholarship awarded to select Nepalese scholars for exceptional academic performance during higher secondary education (+2/High School)

Top 10 Merit Rank, National Physics-Math's Olympiad | IOE, Pulchowk Campus
- Placed in the top 10 nationwide among hundreds of competitive secondary science students in a rigorous quantitative problem-solving championship.

PUBLICATION, CONFERENCE AND MANUSCRIPT

Conference Paper *Published, September 2026*
International Conference on Civil Engineering Innovations and Sustainable Development (CEISD) 2026,
Everest Advances in Science and Technology, Nepal Journal Online

Aryal, S., & Regmi, R. K. (2026). Suspended Sediment Concentration and Load Prediction in the West Rapti River Basin: A Comparative Study of Machine Learning Approaches. Everest Advances in Science and Technology, 2(2), 44-61. https://doi.org/10.3126/east.v2i2.99464

Conference Presentation *26th April, 2026*
CEISD, 2026
Title: Suspended Sediment Concentration and Load Prediction in the West Rapti River Basin: A Comparative Study of Machine Learning Approaches

Manuscript (preprints)

- Statistical Reliability of Gradient-Boosted Ensembles versus ANFIS for Daily Suspended Sediment Concentration Prediction in a Monsoon-Dominated, Highly Variable River Basin
- Optimization of cascade reservoir operation in the Koshi River basin using Non-Dominated Sorting Genetic Algorithm III
- A bootstrap-supported regional framework for design flood estimation in data-scarce Himalayan basins: Evidence from the Narayani Basin, Nepal
- Explainable Machine Learning for Suspended Sediment Concentration Prediction and Catchment Sediment Yield Assessment in a Data-Scarce Himalayan River Basin, Nepal

RESEARCH AND ACADEMIC PROJECTS

M.Sc. Thesis Research "Physics-Guided Machine Learning for Suspended Sediment Prediction"

Department of Hydropower Engineering, Pulchowk Campus

- Benchmarked 10 machine-learning and physics-guided models for daily suspended sediment concentration (SSC) prediction in Nepal's West Rapti River Basin, using 2016–2023 hydro-meteorological records, Optuna hyperparameter tuning, a held-out 2023 test year, and five-fold forward-chaining cross-validation.
- Achieved the strongest overall test-set performance with Random Forest (R^2 = 0.646; RMSE = 439 mg/L; PBIAS = −5.06%), systematically comparing predictive accuracy across model architectures.
- Developed a physics-guided neural network (PGNN) incorporating Yang's sediment-transport capacity theory, a wash-load minimum, and mass conservation; improved test-set R^2 from 0.477 to 0.546 relative to an unconstrained MLP and improved mean cross-validation R^2 from −0.922 to +0.267.
- Revealed a critical model-reliability limitation: all evaluated models performed poorly during high flows ($Q \geq 240.2$ m³/s; high-flow R^2 ranging from +0.04 to −2.08). Identified the dominance of previous-day SSC in predictive importance (48–95%), highlighting the risk of mistaking autoregressive dependence for reliable hydro-meteorological forecasting.

Pre-feasibility Study of Badigad Hydropower Project

- Performed hydrological, hydraulic and topographical feasibility for a proposed peaking run-of-river hydropower development on the Badigad River, and contributed to layout planning and structural sizing of key project components including hydraulic structure, alignment planning, headrace tunnel, and surge shaft
- Prepared a detail energy output analysis with estimated revenue generation.

1D and 2D Hydraulic Modelling of Bagmati River

- Performed and analysis on hydraulic modelling of the Bagmati River corridor for urban flood analysis and land use planning applications.
- Developed and calibrated 1D and 2D HEC-RAS models, producing flood inundation maps for multiple return period scenarios, and analyzed the impact of urban development on flooding around Bagmati River Corridor in Kathmandu.

Preliminary Assessment of Reservoir Sediment Management Strategies

- Investigated sediment management strategies applicable to reservoir hydropower projects in Himalayan contexts
- Applied RESCON 2 software for quantitative assessment of alternative sediment management approaches including flushing, sluicing, and bypassing strategies, for a sustainable use of the reservoir.
- Evaluated long-term reservoir life and sustainability implications of each strategy under varying sediment yield scenarios

Kulekhani Dam Hydrologic Examination

- Conducted seepage analysis of Kulekhani Gravity Dam, Nepal's largest storage hydropower facility
- Undertook a technical site visit to the Kulekhani hydropower station and turbine facility, gaining direct exposure to large-scale hydropower infrastructure

Pre-feasibility and Detail Design of Bhuteni Irrigation Project

- Performed comprehensive hydrological analysis using Department of Hydrology and Meteorology records, including flood frequency analysis and monthly flow estimation using different empirical and theoretical methods.
- Design of irrigation infrastructure including barrage with gated system, frontal intake, triangular canal with rounded bed, settling basin, Sarda fall energy dissipaters, cross regulator, head regulator, Bridge and culvert

PROFESSIONAL EXPERIENCE

Lecturer and Technical Instructor *2026 (currently)*

Institute of Engineering, Pulchowk Campus

- Hydrological modelling using SWAT (Soil Water Assessment Tool), with relation to climate changes scenario
- Hydraulic Modelling (1D and 2D) with addition to Dam Break Analysis using HEC-RAS
- Programming in Hydrology, using Python Programming Languages (prediction model, optimization and data presentation)

Freelance Civil and Hydropower Design Engineer *2025*

Junbesi Khola Cascade Hydropower Project (990 kW), Nepal

- Contributed to civil and structural design of a cascade run-of-river hydropower scheme as part of a collaborative freelance team of three engineers led by a senior engineer with over five years of professional experience.
- Designed hydraulic and structural components including weir, intake structure, headrace pipe, syphon, settling basin, anchor blocks, support piers, turbine housing, and powerhouse layout.

Engineering Intern *2024*

NCER (Nepal Center for Environmental Research), Kathmandu, Nepal)

- Received structured training in applied hydraulic and hydrological modelling using HEC-RAS, HEC-HMS, and EPA SWMM within a professional research and consulting environment
- Developed automation scripts in VBA for processing and organizing hydrological datasets
- Contributed to drainage system design and assisted in the preparation of technical reports

Civil Design Engineer *2023 -2024*

Madi Consultancy Pvt. Ltd., Nepal

- Contributed to irrigation and water supply distribution design projects
- Supported technical documentation, site assessment, and preliminary hydraulic design activities

TECHNICAL TRAINING AND CERTIFICATIONS

Hydraulic Design Software Training

- Intensive training in HEC-RAS, HEC-ResSim, and HEC-HMS for hydraulic routing, reservoir simulation, and rainfall-runoff modelling
- Training in SWAT (Soil and Water Assessment Tool) for catchment-scale hydrological simulation and water quality modelling
- Machine Learning for Hydrological Applications self-directed graduate study with applied implementation in MSc thesis research
- Google Earth Engine remote sensing and geospatial data processing for hydrological applications

HOBBIES AND INTERESTS

- Hiking, Trekking and Camping
- Football (1st year football cup winner, futsal cup runner up, KMC football tournament winner, Reliance annual feast runner-up)
- Math quizzes and Olympiads
- Technology and Innovations
- Cricket (annual Cricket cup winner)

TECHNICAL SKILLS AND COMPETENCIES

Hydrological Modelling	HEC-HMS, HEC-RAS (1D/2D), HEC-ResSim, SWAT, EPA SWMM, RESCON 2
Machine Learning	Regression, ensemble methods (Random Forest, Gradient Boosting), neural networks; scikit-learn, pandas, NumPy, Matplotlib
GIS and Remote Sensing	ArcGIS, QGIS, Google Earth Engine
CAD and Design	AutoCAD, Civil 3D

Programming	Python, VBA, JavaScript, HTML, C, FORTRAN
CFD Simulation	OpenFOAM (foundational level)
Scientific Writing	Technical report preparation, academic manuscript drafting, LaTeX
Office and Data Tools	MS Excel (advanced), MS Word, MS PowerPoint

LANGUAGE PROFICIENCY

- Nepali — Native proficiency
- English — Professional working proficiency (written and spoken); actively developing academic writing competency

PROFESSIONAL AFFILIATIONS

- Member, Nepal Engineering Council (NEC): Licensed Civil Engineer
- Graduate, Pulchowk Campus Alumni Network, Institute of Engineering, Tribhuvan University

REFERENCES

Academic and professional references are available upon request. Referees include faculty members from the Department of Hydropower Engineering, Pulchowk Campus, and senior professionals from the hydropower engineering sector in Nepal.